PRANAV SANGANI

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EDUCATION

Purdue University, *Bachelor of Science in Computer Science – Junior* May 2028
Certificate in Entrepreneurship and Innovation

CS Coursework: Data Structures and Algorithms, Problem Solving and Object-Oriented Programming, Computer Architecture, Programming in C, Foundations of Computer Science, Python Programming, Tools (Unix, Git)

Entrepreneurship Coursework: Introduction to Entrepreneurship and Innovation, Marketing and Management for New Ventures, Internship and Career Preparation Seminar

Clubs: Computer Science Club, Boomi (Indian Student Association), Chess Club

EXPERIENCE

Axcellion, *Software Engineer Intern* May 2026 – Present

- Built an end-to-end RAG pipeline in Python on AWS (S3, EC2) that parses and embeds **80K+ pages** of FDA drug documents into a Neo4j knowledge graph.
- Raised search accuracy by **over 50%** by engineering hybrid retrieval on AWS, then extended it to EU drug data.
- Unified FDA and EU retrieval on shared reference data and a common reranking stage, standardizing search across **240K+ document chunks**.
- Led development of a federated search API in Python that answers one query across FDA and EU data sources in **~0.3 seconds**, with **zero hallucinated answers** in audit.
- Benchmarked several parsers and embedding models, choosing a stack that cut costs **~30x** at **99.5% accuracy**.
- Reorganized and consolidated company GitHub repositories, unifying **3 production pipelines** into one codebase.

Aavalar Consulting, *Software Engineer Intern* June 2026 – Sept 2026

- Co-built SHERPA, an AI assistant on the Claude Agent SDK and MCP that answers employee questions from company documents with citations, integrated with Microsoft Teams and SharePoint.
- Designed permission-aware document search in PostgreSQL with **row-level security**, so users only see files they're authorized to access, backed by **220+ automated tests** in CI on GitHub.
- Built a multi-agent evaluation pipeline in Python to measure answer accuracy, refusals, and access control.

DEEPSEAS (Drone Enabled Environment Patrol & Surveillance Edge AI System) Oct 2020 – June 2024

- Co-developed a patent-pending underwater buoy that detects illegal fishing in Marine Protected Areas.
- Ran AI audio and vision models on a Jetson Nano edge device, trained on a curated dataset to identify vessel types.

GCL AI Summit - *Capital One Hall* Sept 2022 – Aug 2023

- Co-led business operations for a charity AI summit for Title I students, featuring speakers from Meta and NVIDIA.
- Helped secure sponsorships from Google, Meta, Intel AI, and AWS; NBC-4 promotion drew **around 300 students**.

SKILLS

Languages: Python, Java, C, SQL, R

Frameworks & Tools: AWS (S3, EC2), Claude Agent SDK, Claude Code, MCP, PostgreSQL, Git, GitHub, Linux

Certifications: Claude Certified Architect – Foundations *Anthropic | July 2026*